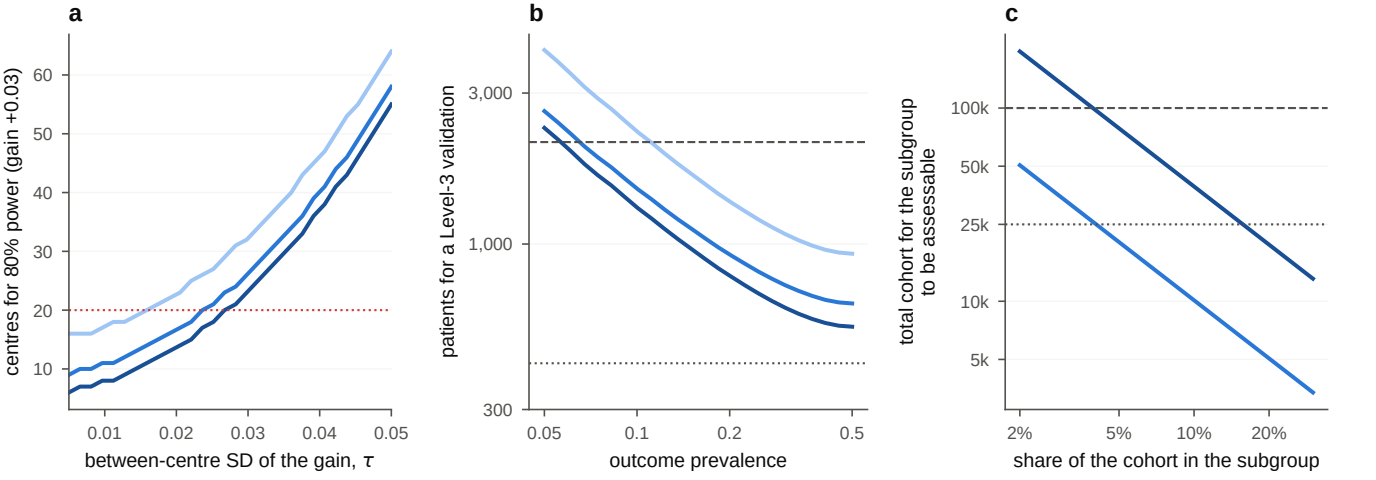
a
centres for 80% power (gain +0.03)
60
50
40
30
20
10
0.01
0.02
0.03
0.04
0.05
between-centre SD of the gain, τ
b
patients for a Level-3 validation
3,000
1,000
300
0.05
0.1
0.2
0.5
outcome prevalence
c
total cohort for the subgroup to be assessable
100k
50k
25k
10k
5k
2%
5%
10%
20%
share of the cohort in the subgroup
150 patients per centre
300 patients per centre
600 patients per centre
gate minimum (20)
AUROC 0.70
AUROC 0.75
AUROC 0.80
one centre, 3 years
20 centres, 1 year
sensitivity gap 0.10
sensitivity gap 0.05
25,000-patient extract
100,000-patient extract